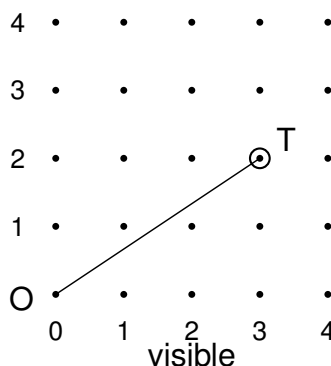
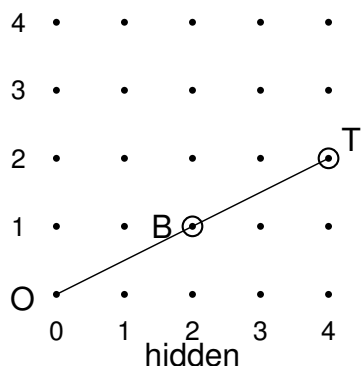
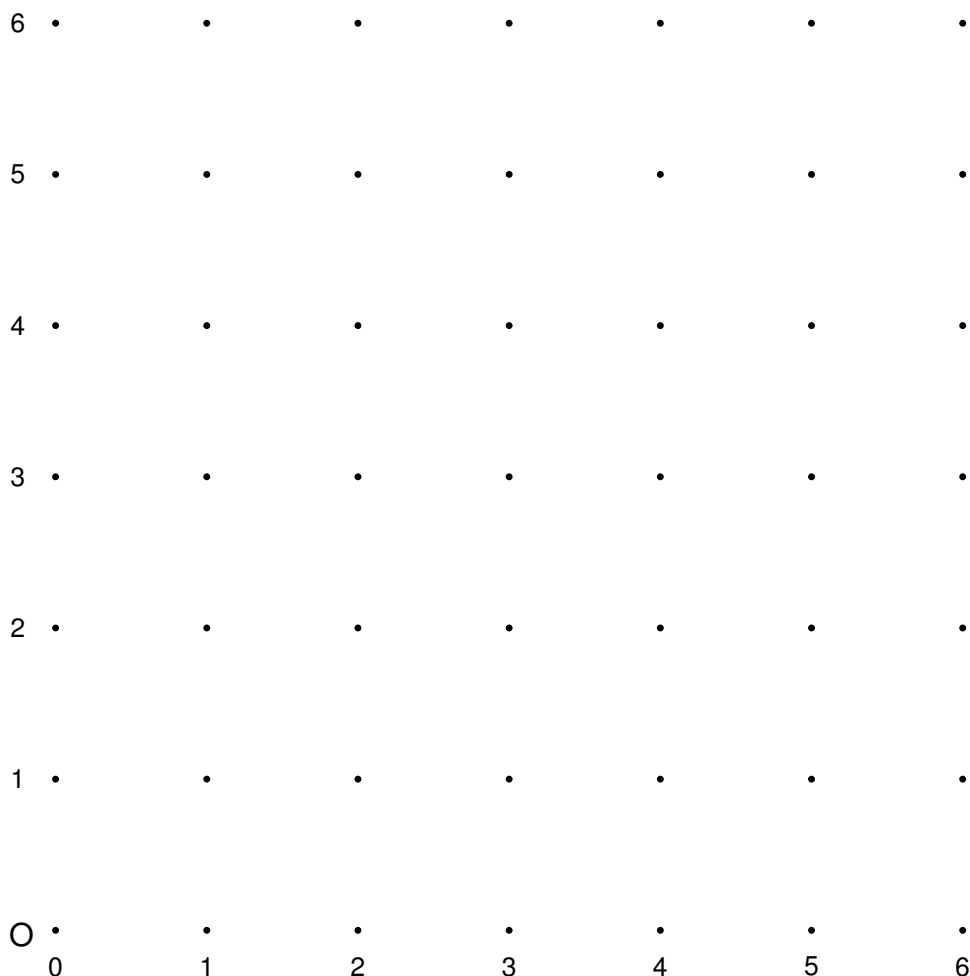


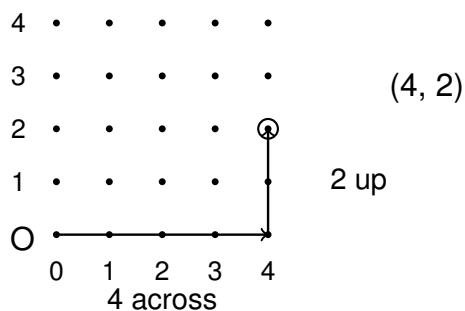
Stretch a thread from O to another dot T. A dot exactly on the thread between O and T blocks T. B marks a blocker. Keep O fixed while you test.



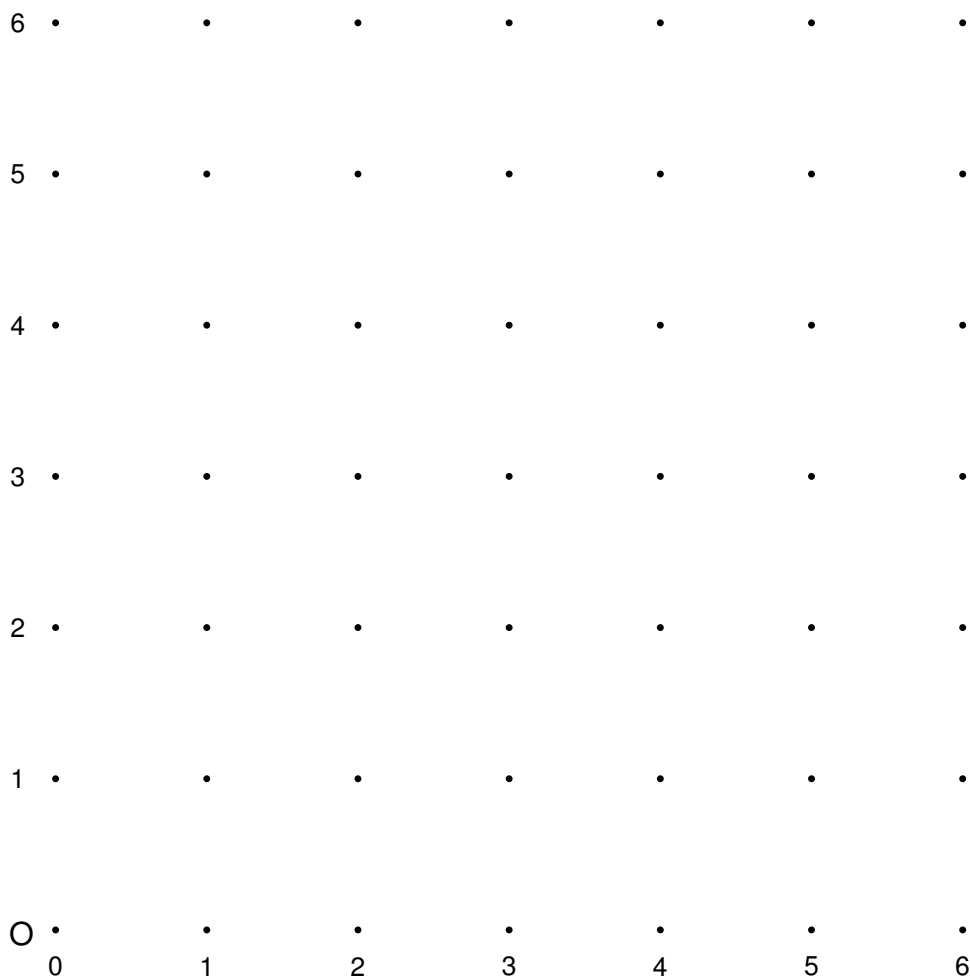
Problem 1: Find three visible targets and three hidden targets with different numbers of blockers. Mark each blocker.



(4, 2) means 4 across and 2 up from O.



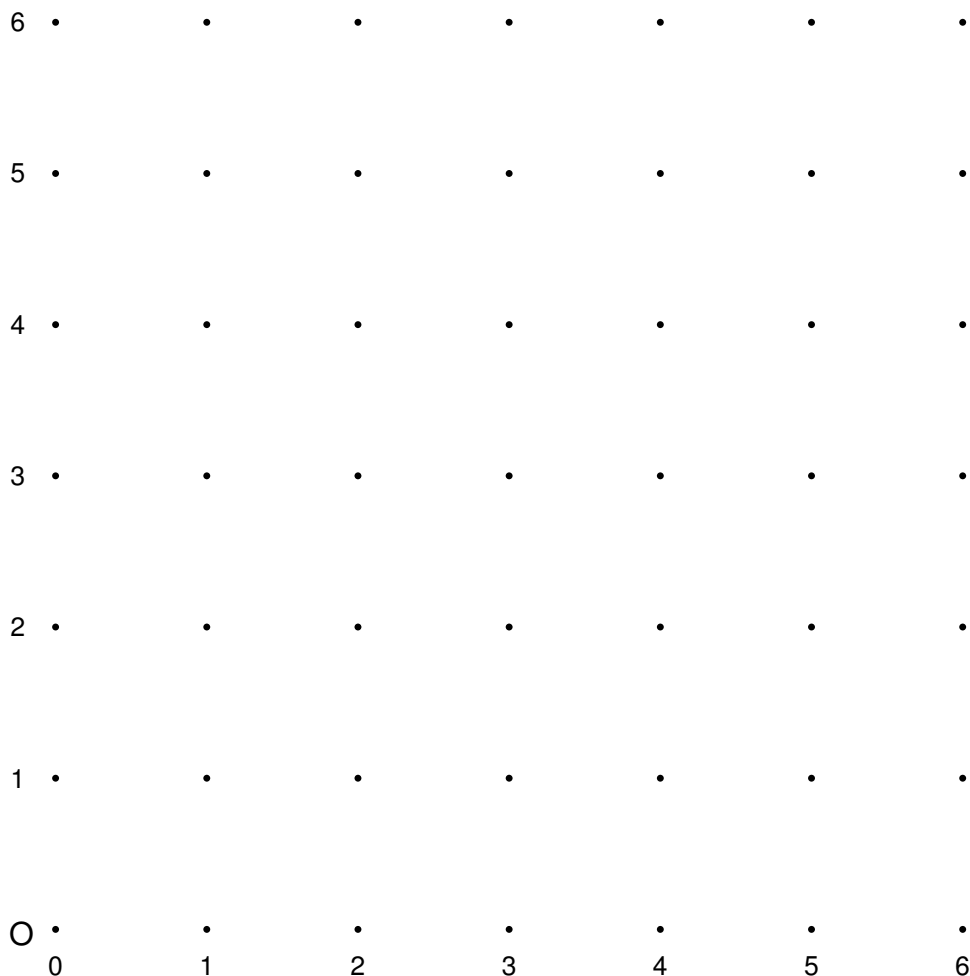
Problem 2: For each target, find the first dot after O on its line and every blocker:
 (6, 4), (6, 3), (5, 3), (6, 6), (6, 0) and (0, 6).



Problem 3: Predict the first dot and number of blockers for $(12, 8)$, $(15, 10)$, $(21, 14)$, $(25, 15)$ and $(13, 8)$. Explain a rule that works for any target with whole-number coordinates.

Problem 4: A target is (a, b) , with a and b positive whole numbers. Explain why a common divisor larger than 1 always produces a blocker. Does every blocker produce a common divisor?

Problem 5: List all the targets (a, b) whose two coordinates are each from 1 through 6 that have the same first visible dot as $(6, 4)$. Repeat for $(6, 3)$ and $(6, 6)$. Explain why no target can belong to two different groups.



Problem 6: Let d be the greatest whole number that divides both coordinates of a target. How does d determine the first dot after O and the number of blockers? Explain why there can be no extra dots between the ones your rule finds.

Problem 7: Find an entire row of visible dots, an infinite collection of hidden dots, and a target with exactly 99 blockers. Explain each choice.
